

Moving Both Left and Right

Now that the ship can move continuously to the right, adding movement to the left is straightforward. Again, we'll modify the Ship class and the `_check_events()` method. Here are the relevant changes to `__init__()` and `update()` in Ship:

```
ship.py    def __init__(self, ai_game):
            --snip--
            # Movement flags; start with a ship that's not moving.
            self.moving_right = False
            self.moving_left = False

            def update(self):
                """Update the ship's position based on movement flags."""
                if self.moving_right:
                    self.rect.x += 1
                if self.moving_left:
                    self.rect.x -= 1
```

In `__init__()`, we add a `self.moving_left` flag. In `update()`, we use two separate `if` blocks, rather than an `elif`, to allow the ship's `rect.x` value to be increased and then decreased when both arrow keys are held down. This results in the ship standing still. If we used `elif` for motion to the left, the right arrow key would always have priority. Using two `if` blocks makes the movements more accurate when the player might momentarily hold down both keys when changing directions.

We have to make two additions to `_check_events()`:

```
alien_invasion.py def _check_events(self):
                    """Respond to keypresses and mouse events."""
                    for event in pygame.event.get():
                        --snip--
                        elif event.type == pygame.KEYDOWN:
                            if event.key == pygame.K_RIGHT:
                                self.ship.moving_right = True
                            elif event.key == pygame.K_LEFT:
                                self.ship.moving_left = True

                        elif event.type == pygame.KEYUP:
                            if event.key == pygame.K_RIGHT:
                                self.ship.moving_right = False
                            elif event.key == pygame.K_LEFT:
                                self.ship.moving_left = False
```

If a `KEYDOWN` event occurs for the `K_LEFT` key, we set `moving_left` to `True`. If a `KEYUP` event occurs for the `K_LEFT` key, we set `moving_left` to `False`. We can use `elif` blocks here because each event is connected to only one key. If the player presses both keys at once, two separate events will be detected.

When you run *alien_invasion.py* now, you should be able to move the ship continuously to the right and left. If you hold down both keys, the ship should stop moving.